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# Problem Set 3

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PY 451 Section A1

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## 1 Representations and Basis States

Even though our space is infinite-dimensional, we can isolate  $v_1, v_2$  to consider a 2-dimensional subspace. The most general possible choice in this subspace, as illustrated by equation (1.1), is

$$|1\rangle = \cos \theta e^{i\phi_1} v_1 + \sin \theta e^{i\phi_2} v_2,$$

$$|2\rangle = \sin \theta e^{i\phi_3} v_1 - \cos \theta e^{i(\phi_3 + \phi_1 - \phi_2)} v_2.$$

Idfk what to do for b

## 2 Operators

1. h
2. The question is wrong, since  $h$  is not continuous, and thus  $h \notin V$  so  $Ah$  is not defined. The same is true for  $g$ . I will ignore this inconvenient truth to give the intended answer. Note that  $h(x)g(x) = 0$  (except possibly on  $\{1/3, 2/3\}$  depending on how the drawing is interpreted, but this set has lebesgue measure 0 so the following conclusion still holds). It follows that

$$\int_0^1 h(y)g(y)g(x) \, dy = 0.$$

Therefore,

$$(Ah)(x) = \int_0^1 h(y)^2 h(x) \, dy + \int_0^1 h(y)v(y)v(x) \, dx.$$

Note that  $v(x) = \frac{1}{3}h(x)$ . It follows that

$$(Ah)(x) = \int_0^1 h(y)^2 h(x) \, dy + \int_0^1 \frac{1}{9} h(y)^2 h(x) \, dy = \frac{10}{9} h(x) \int_0^1 h(y)^2 \, dy.$$

Also note that since  $0^2 = 0$  and  $1^2 = 1$ , we have that  $h(x)^2 = h(x)$ , so

$$(Ah)(x) = \frac{10}{9} h(x) \int_0^1 h(y) \, dy = \frac{20}{27} h(x).$$

Computing  $(Ag)(x)$  is more simple, since the first and last integrals vanish. We have

$$(Ag)(x) = g(x) \int_0^1 g(y)^2 \, dy = g(x) \int_{1/3}^{2/3} dx = \frac{1}{3} g(x).$$

3. The intended solution is  $20/27$  with  $h(x)$  as an eigenfunction,  $1/3$  with  $g(x)$  as an eigenfunction, and any scalar multiple of these functions. Note that the image of  $A$  must be a subset of the span of  $\{h, g\}$  since  $v = 1/3h$  is in this span, and thus these are clearly the only possible linearly independent eigenspaces other than the  $0$  eigenspace.

The question is once again worded wrong, however, and if we take  $V$  to actually be the space of continuous functions and  $A$  to be a map  $V \rightarrow \mathbb{R}^{[0,1]}$ , then the answer becomes only  $0$  as an eigenvalue, since clearly any continuous eigenfunction  $f$  of a nonzero eigenvalue must be constant, and plugging a nonzero constant into the integrals implies the contradiction that  $1/3 = 10/9$ . The eigenfunctions of the  $0$  eigenvalue are functions  $f \in V$  such that

$$\int_0^{1/3} f(x) dx = - \int_{2/3}^1 f(x) dx, \text{ and } \int_{1/3}^{2/3} f(x) dx = 0.$$

This will cause  $Af = 0$ .

### 3 Born's Rule and Expectation Values

1. Recall:

$$S_{\vec{n}} = \begin{pmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{pmatrix},$$

and the eigenvectors associated with eigenvalues  $\hbar/2, -\hbar/2$  are

$$|\psi_1\rangle = \cos \frac{\theta}{2} |+\rangle + \sin \frac{\theta}{2} |-\rangle, \quad |\psi_2\rangle = \sin \frac{\theta}{2} |+\rangle - \cos \frac{\theta}{2} |-\rangle,$$

respectively. We have

$$\mathcal{P}_{\hbar/2} = |\langle + | \psi_1 \rangle|^2 = \left| \cos \frac{\theta}{2} \langle + | + \rangle + \sin \frac{\theta}{2} \langle + | - \rangle \right|^2 = \left| \cos \frac{\theta}{2} \right|^2.$$

Similarly, we have

$$\mathcal{P}_{-\hbar/2} = |\langle + | \psi_2 \rangle|^2 = \left| \sin \frac{\theta}{2} \langle + | + \rangle - \cos \frac{\theta}{2} \langle + | - \rangle \right|^2 = \left| \sin \frac{\theta}{2} \right|^2.$$

The former is the probability of measuring  $\hbar/2$ , since  $|\psi_1\rangle$  is the corresponding eigenvector, and the latter is the probability of finding  $-\hbar/2$  for the same reason. As a quick sanity check,

$$\left| \sin \frac{\theta}{2} \right|^2 + \left| \cos \frac{\theta}{2} \right|^2 = \sin^2 \frac{\theta}{2} + \cos^2 \frac{\theta}{2} = 1.$$

2. (a)

$$\begin{aligned} \langle + | S_{\vec{n}} | + \rangle &= \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ &= \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix} \\ &= \frac{\hbar}{2} \cos \theta. \end{aligned}$$

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(b)

$$\langle S_{\vec{n}} \rangle = \sum_i \lambda_i \mathcal{P}_i = \frac{\hbar}{2} \cos^2 \frac{\theta}{2} - \frac{\hbar}{2} \sin^2 \frac{\theta}{2} = \frac{\hbar}{2} \left( \cos^2 \frac{\theta}{2} - \sin^2 \frac{\theta}{2} \right) = \frac{\hbar}{2} \cos \theta.$$

3. (a) Note that

$$\mathcal{P}_{\hbar/2} = |\langle \psi_1 | \psi_1 \rangle|^2 \equiv 1.$$

It follows that  $\mathcal{P}_{-\hbar/2} = 0$  by classical probability theory. Therefore,  $|\psi_1\rangle$  and  $|\psi_2\rangle$  are orthogonal states.

(b) i.

$$\begin{aligned} \langle \psi_1 | S_{\vec{n}} | \psi_1 \rangle &= \frac{\hbar}{2} \begin{pmatrix} \cos \frac{\theta}{2} & \sin \frac{\theta}{2} \end{pmatrix} \begin{pmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{pmatrix} \begin{pmatrix} \cos \frac{\theta}{2} \\ \sin \frac{\theta}{2} \end{pmatrix} \\ &= \frac{\hbar}{2} \begin{pmatrix} \cos \frac{\theta}{2} & \sin \frac{\theta}{2} \end{pmatrix} \begin{pmatrix} \cos \theta \cos \frac{\theta}{2} + \sin \theta \sin \frac{\theta}{2} \\ \cos \frac{\theta}{2} \sin \theta - \sin \frac{\theta}{2} \cos \theta \end{pmatrix} \\ &= \frac{\hbar}{2} \left( \cos \theta \cos^2 \frac{\theta}{2} + \sin \theta \cos \frac{\theta}{2} \sin \frac{\theta}{2} + \cos \frac{\theta}{2} \sin \frac{\theta}{2} \sin \theta - \sin^2 \frac{\theta}{2} \cos \theta \right) \\ &= \frac{\hbar}{2} \left( \cos \theta \left( \cos^2 \frac{\theta}{2} - \sin^2 \frac{\theta}{2} \right) + 2 \sin \theta \cos \frac{\theta}{2} \sin \frac{\theta}{2} \right) \\ &= \frac{\hbar}{2} (\cos^2 \theta + \sin^2 \theta) \\ &= \frac{\hbar}{2}. \end{aligned}$$

ii.

$$\langle S_{\vec{n}} \rangle = \sum_i \lambda_i \mathcal{P}_i = \frac{\hbar}{2} 1 - \frac{\hbar}{2} 0 = \frac{\hbar}{2}.$$